In earlier classes, using the above relations between roots and coefficients we constructed a quadratic equation, having α and β as roots. In fact, such an equation is given by (1). For instance, a quadratic equation whose roots are 3 and 4 is given by $x^2-7x+12=0$.

Further we construct new polynomial equations whose roots are functions of the roots of a given polynomial equation; in this process we form a new polynomial equation without finding the roots of the given polynomial equation. For instance, we construct a polynomial equation by increasing the roots of a given polynomial equation by two as given below.

Example 3.1

If α and β are the roots of the quadratic equation $17x^2+43x-73=0$, construct a quadratic equation whose roots are $\alpha+2$ and $\beta+2$.

Solution

Since α and β are the roots of $17x^2+43x-73=0$, we have $\alpha+\beta=\frac{-43}{17}$ and $\alpha\beta=\frac{-73}{17}$.

We wish to construct a quadratic equation with roots $\alpha+2$ and $\beta+2$. Thus, to construct such a quadratic equation, calculate,

$$\text{the sum of the roots} = \alpha+\beta+4 = \frac{-43}{17}+4 = \frac{25}{17} \quad \text{and}$$

$$\text{the product of the roots} = \alpha\beta+2(\alpha+\beta)+4 = \frac{-73}{17}+2\left(\frac{-43}{17}\right)+4 = \frac{-91}{17}.$$

Hence a quadratic equation with required roots is $x^2-\frac{25}{17}x-\frac{91}{17}=0$.

Multiplying this equation by 17, gives $17x^2-25x-91=0$

which is also a quadratic equation having roots $\alpha+2$ and $\beta+2$.

Example 3.2

If α and β are the roots of the quadratic equation $2x^2-7x+13=0$, construct a quadratic equation whose roots are α^2 and β^2.

Solution

Since α and β are the roots of the quadratic equation, we have $\alpha+\beta=\frac{7}{2}$ and $\alpha\beta=\frac{13}{2}$.

Thus, to construct a new quadratic equation,

$$\text{Sum of the roots} = \alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta=\frac{-3}{4}.$$

$$\text{Product of the roots} = \alpha^2\beta^2=(\alpha\beta)^2=\frac{169}{4}$$

Thus a required quadratic equation is $x^2+\frac{3}{4}x+\frac{169}{4}=0$. From this we see that

$$4x^2+3x+169 = 0$$

is a quadratic equation with roots α^2 and β^2.

Remark

In Examples 3.1 and 3.2, we have computed the sum and the product of the roots using the known $\alpha+\beta$ and $\alpha\beta$. In this way we can construct quadratic equation with desired roots, provided the sum and the product of the roots of a new quadratic equation can be written using the sum and the product of the roots of the given quadratic equation. We note that we have not solved the given equation; we do not know the values of α and β even after completing the task.

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